

DYNAMIC TREASURE HUNT

Product Requirements Document | Command-Line Adventure Game

1. Problem Statement & Description

Dynamic Treasure Hunt is a command-line adventure game in which a player explores a map to find a hidden treasure. The player starts at CAMP and must solve clues to determine where to travel next. The player cannot choose a destination directly; the answer to each clue decides the next location. The player must manage energy, chances, hints and score while some roads may temporarily close or reopen. The treasure is randomly selected and hidden. The player wins by reaching it before the expedition ends.

2. Interface

The interface examples below show the terminal input and output used by the game.

Startup Screen

```

      ◆ DYNAMIC TREASURE HUNT ◆
    CLUE • CHOOSE • EXPLORE • SURVIVE

  A colorful adventure of clues, routes and hidden treasure

      ◆ YOUR EXPEDITION BEGINS HERE ◆
    Manage energy • chances • score

    Press ENTER to continue...
```

Main Menu

```

=====
                        TREASURE HUNT
=====
1. START NEW EXPEDITION
2. VIEW MAP
3. HOW TO PLAY
4. EXIT
=====
Choose an option: 1
```

Difficulty Selection

```

=====
                        CHOOSE DIFFICULTY
=====
1. Easy
2. Medium
3. Hard
Choose difficulty: 1

=====
                        EXPEDITION READY
=====
Difficulty selected: EASY

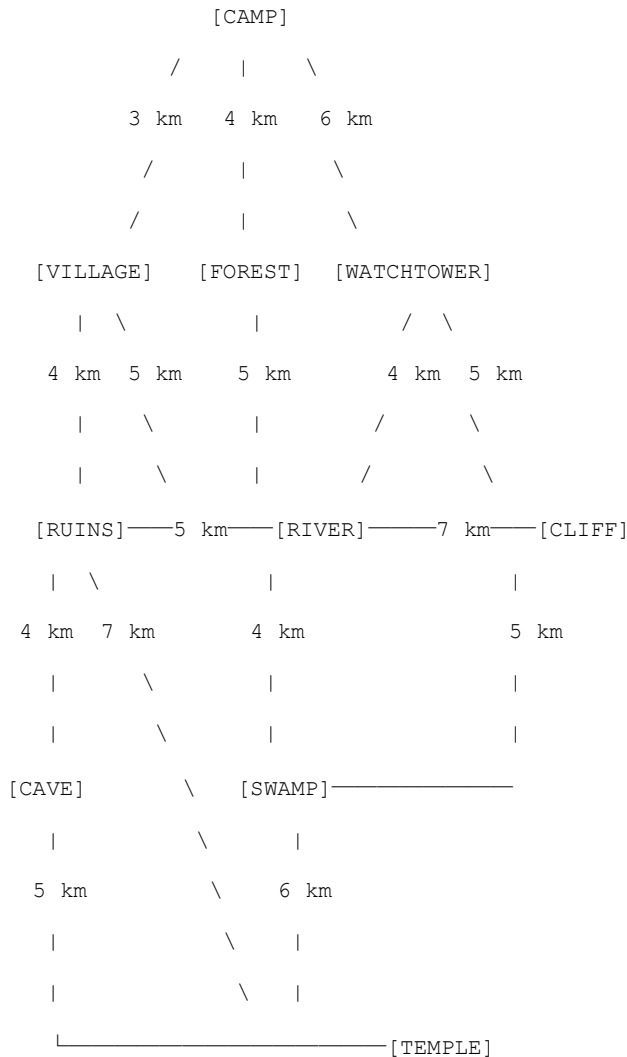
STARTING RESOURCES
```

⚡ Energy : 42
🎯 Chances : 16
💡 Hints : 5

Your treasure location is hidden.
Follow the clues, use hints when needed, and manage your resources.

Press ENTER to start the expedition...

Map



Clue Input / Output

CLUE

EASY CLUE: You are at CAMP. Which location can you reach directly
through an OPEN route?

1. FOREST
2. VILLAGE
3. WATCHTOWER

H = Hint | M = Map | S = Status | Q = Quit

Your choice: 1

✓ Correct!

Points gained: 50

The map confirms that CAMP connects to FOREST (4 km, 5 energy).

CLUE SOLVED — FOLLOWING THE CLUE TO FOREST

✓ You moved to FOREST

Distance travelled: 4 km

Energy used for route: 5

Energy remaining: 37

Chances remaining: 15

Hint Input / Output

H = Hint | M = Map | S = Status | Q = Quit

Your choice: H

Energy lost for using hint: 2

Energy remaining: 40

Hint: VILLAGE is not the correct answer.

Status Input / Output

H = Hint | M = Map | S = Status | Q = Quit

Your choice: S

CURRENT STATUS

Location: FOREST

Energy: 37

Chances: 15

Hints: 4

Score: 100

Wrong Answer Input / Output

H = Hint | M = Map | S = Status | Q = Quit

Your choice: 2

X WRONG ANSWER!

Energy lost: 1

Chance lost: 1

Score lost: 15

Energy remaining: 36

Chances remaining: 14

Try the clue again.

3. Functional Requirements

FR-01: Start the Game

The system shall display the startup screen and main menu. The player can start a new expedition, view the map, read the instructions, or exit.

FR-02: Select Difficulty

The system shall allow the player to select Easy, Medium, or Hard. The selected level controls starting energy, chances, hints, points, wrong-answer penalties, and initially closed roads.

FR-03: Create a Hidden Treasure

At the start of each expedition, the system shall randomly select one location other than CAMP as the treasure location. The location shall remain hidden.

FR-04: Start at CAMP

Every new expedition shall start at CAMP with 100 score and 0 km travelled.

FR-05: Display the Map

The system shall show the 10 locations, connected roads, road distances, the player's current location, and any closed roads. The treasure shall not be shown.

FR-06: Generate Clues

The system shall generate clues from the player's current location and open routes. The game shall use direct-route clues, distance/comparison clues, and two-step map clues. Difficulty shall affect clue difficulty and wording.

FR-07: Answer a Clue

The player shall select one of the displayed answers. A correct answer awards the difficulty-based points and determines the next destination.

FR-08: Penalize a Wrong Answer

A wrong answer without a hint shall reduce energy, score, and one chance according to the selected difficulty. The same clue shall remain active for another attempt. Score shall not go below zero. If the player gives a wrong answer after using a hint, the system shall apply double the normal score penalty, while no additional energy or chance is lost.

FR-09: Use Hints

The player can press H to remove one incorrect option. One hint is consumed each time, and energy is reduced by twice the normal wrong-answer energy penalty for the selected difficulty. Using a hint does not reduce score or chances. If no hints remain, the system shall report that none are available. If the hint reduces energy to zero or below, the expedition shall end because the player does not have enough energy to make the next move.

FR-10: View the Map During a Clue

The player can press M to view the map and then return to the same clue.

FR-11: View Current Status

The player can press S to view current location, energy, chances, hints, and score.

FR-12: Move Automatically

After a correct clue answer, the system shall automatically move the player to the answer location. The route's energy cost is deducted, one chance is used, and its distance is added to total travelled distance.

FR-13: Change Routes Dynamically

During the expedition, the system may close an open route or reopen a closed route. A route change shall not make the hidden treasure unreachable.

FR-14: Avoid Recent Clue Repetition

The system shall avoid repeating previously generated clues by creating a clue ID from the question and its options. It shall check the clue against the current game's used questions and the most recent five completed games while the program is running. If a clue has already been used, the system shall try to generate another clue.

FR-15: End the Expedition

The expedition shall end when the player reaches the treasure, runs out of energy or chances, quits, cannot make the required next move, has no open route, or no new clue is available. The system shall show the appropriate result and reason.

Difficulty and Resource Rules

Difficulty	Energy	Chances	Hints	Points / Correct Answer	Wrong Energy	Wrong Score
Easy	42	16	5	50	1	15
Medium	32	13	4	80	2	20
Hard	27	11	3	120	3	30

Hint and Post-Hint Penalty Rules

Difficulty	Hint Energy Cost	Score Lost for Wrong Answer After Hint
Easy	-2 energy	-30 score
Medium	-4 energy	-40 score
Hard	-6 energy	-60 score

Using a hint does not directly reduce score or chances. If the player then gives a wrong answer, only the double score penalty shown above is applied; no additional energy or chance is lost.

Test Cases

Test Case	Input / Situation	Expected Output	FR Covered
TC1 – Start the Game	At the main menu, enter `1`.	The system displays CHOOSE DIFFICULTY with `1. Easy`, `2. Medium`, `3. Hard`, and `Choose difficulty:`.	FR-01
TC2 – Select Medium	At difficulty selection, enter `2`.	The system displays EXPEDITION READY, `Difficulty selected: MEDIUM`, and starting resources: Energy: 32, Chances: 13, Hints: 4. It also displays `Your treasure location is hidden.`	FR-02
TC3 – Hidden Treasure and Starting Location	Start a new expedition and press ENTER to begin.	The system states that the treasure location is hidden and the expedition starts at CAMP. Initial score is 100 and travelled distance is 0 km.	FR-03, FR-04
TC4 – Display Map	During the expedition, enter `M`.	The system displays the treasure-hunt map with locations, connected routes, distances and currently closed routes. The player can return to the current clue.	FR-05
TC5 – Generate Clue	Start/continue an expedition until a clue is displayed.	The system displays a clue based on the player's current location and available routes, with numbered route options. The clue varies according to the selected difficulty.	FR-06

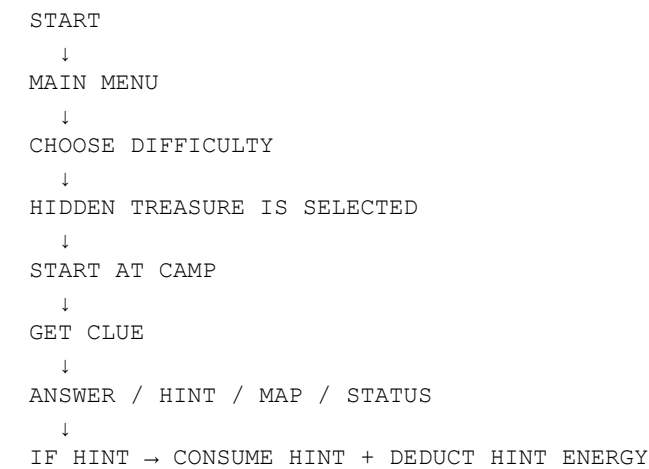
TC6 – Correct Route and Automatic Movement	Select the route number that satisfies the displayed clue. Example observed: enter `1` for `FOREST`.	The system displays `✓ Correct!`, awards the difficulty-based points, shows the explanation, follows the clue to the selected destination, displays movement details, deducts route energy and updates Chances and distance.	FR-07, FR-12
TC7 – Wrong Answer Without Hint	Select an incorrect route without using a hint.	The system displays `X WRONG ANSWER!`, applies the selected difficulty's normal energy, chance and score penalties, updates the resources, and displays `Try the clue again.`	FR-08
TC8 – Use Hint	During a clue, enter `H`.	The system consumes one hint, removes one incorrect option, and deducts the configured hint energy cost. Score and chances are not reduced by using the hint.	FR-09
TC9 – Wrong Answer After Hint	During a clue, enter `H` and then select an incorrect answer.	The system applies double the normal score penalty. No additional energy or chance is lost for the wrong answer. The clue remains active if resources are still available.	FR-08, FR-09
TC10 – View Map During Clue	While a clue is active, enter `M`.	The map is displayed and, after returning, the same clue remains active for the player to answer.	FR-10
TC11 – View Current Status	During a clue, enter `S`.	The system displays CURRENT STATUS with Location, Energy, Chances, Hints, and Score.	FR-11
TC12 – Dynamic Route Event	Continue the expedition with successful moves until a dynamic event occurs.	The system displays an event such as `A blocked route has reopened!` and updates the route state accordingly. The route change does not make the hidden treasure unreachable.	FR-13
TC13 – Find Treasure	Continue solving clues until the player reaches the hidden treasure.	The system ends the expedition and displays the treasure-found/final result, including the final expedition information.	FR-15
TC14 – Resource Exhaustion	Continue the expedition until an ending condition such as energy or chances being exhausted occurs.	The system ends the expedition and displays the appropriate failure reason and final expedition information.	FR-15
TC15 – Quit Expedition	During a clue, enter `Q`.	The system immediately ends the expedition, displays the reason for quitting and the final	FR-15

		expedition information, and returns to the main menu.	
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Edge Cases

Input / Situation	Expected Behaviour
Invalid main-menu choice	Enter an invalid value such as `sdfghj`. The system displays the invalid-choice message and shows the main menu again.
Invalid difficulty input	At difficulty selection, enter an invalid value. The system displays the invalid-difficulty message and returns to difficulty selection.
Invalid route number	Enter a route number that is not one of the displayed choices. The system displays the invalid route/choice message and does not move the player.
Unsupported command	During a clue, enter an unsupported value such as `ertyuij`. The system displays `Invalid command.` and keeps the player at the current clue.
Closed path selected	Select a route that is currently closed. The system displays the path-closed warning and does not move the player.
Insufficient energy	Select a route requiring more energy than currently available. The system displays the insufficient-energy message and does not move the player.
Wrong challenge answer	Select an incorrect route. The system displays `X WRONG ANSWER!`, applies the configured penalties and shows the clue again.
No hints remaining	Enter `H` when no hints are available. The system displays `No hints remaining.` and does not provide another hint.
Hint exhausts energy	Use a hint when the player's remaining energy is equal to or less than the hint energy cost. The system deducts the hint energy cost, and if energy reaches zero or below, the expedition ends because the player does not have enough energy for the next move. The hint does not reduce chances.
Energy exhausted	Continue until energy reaches zero. The expedition ends and displays the applicable failure reason and final information.
Chances exhausted	Continue giving wrong answers until chances reach zero. The expedition ends and displays the applicable failure reason and final information.
No usable route remains	When there is no usable route for the next move, the expedition ends with the applicable failure reason.
Treasure reached	When the player reaches the hidden treasure, the expedition stops and the final result is displayed.

Complete Game Flow



↓
ANSWER CLUE
↓
CORRECT → SCORE + MOVEMENT
OR
WRONG → APPLY PENALTY
↓
POSSIBLE ROUTE EVENT
↓
NEXT CLUE
↓
REPEAT
↓
TREASURE FOUND → VICTORY
OR
RESOURCE / OTHER END CONDITION → EXPEDITION ENDED